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CLINICAL FEATURE
ORIGINAL RESEARCH

Thursday Night Football's impact on all-cause injuries in NFL players during 2012 – 2017

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ABSTRACT

Objectives: This study sought to determine whether shortened recovery periods between regularly scheduled Sunday NFL games and Thursday Night Football games significantly increased the incidence of injury in NFL players.

Methods: NFL injury reports and injury reserve data were collected for every NFL player, on all 32 NFL teams, for each week during the regular season, for the 2012–2013, 2013–2014, 2014–2015, 2015–2016, and 2016–2017 NFL seasons. Injuries were defined as any reported injury, not previously documented in an injury report prior to said injury. Calculated injury rates per 1000 athletic exposures for Sunday and Monday night games versus Thursday night games was used to generate relative risk of injury using 95% confidence intervals.

Results: The all-cause injury rate during NFL Sunday and Monday Night Football games was found to be 7,598 per 1000 athletic exposures, while the all-cause injury rate during Thursday Night Football games was found to be 6,072 per 1,000 athletic exposures. The relative risk of injury during Thursday Night Football games was calculated to be 0.97 compared to Sunday and Monday night games. Therefore, the rate of injury during Thursday Night Football games was significantly less than the rate of injury during Sunday and Monday night games, despite the lack of additional recovery time.

Conclusion: This study suggests that eliminating Thursday Night Football is unlikely to improve the statistical injury rate among NFL players.

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KEYWORDS
American football; epidemiology; schedule; sports medicine

Introduction

The National Football League (NFL) launched Thursday Night Football programming as a means to capitalize on its successful ratings, adding an additional weekly football game played on Thursday evenings. Like Monday Night Football, Thursday night football has been popular with the NFL fan base, likely because of its national TV spotlight. Thursday night football has been especially beneficial for NFL teams in smaller media markets that do not consistently receive national broadcasting. Despite Thursday Night Football's popularity with NFL fans, NFL players have publicly opposed Thursday night games citing a lack of proper recovery time [1]. Traditionally NFL football teams play every Sunday, thus leaving six days of rest between games. Teams that play Thursday nights, though, only get three days of rest. Prominent NFL players have criticized the NFL for scheduling Thursday night games [1].

Prior studies have shown connections between lack of rest and injury [2]. Wang et al. demonstrated that muscle stiffness is significantly increased in the setting of muscle fatigue [3]. This is of concern because Witvrouw et al. found that increased tightness of the hamstring or quadriceps muscles is at a higher risk for a subsequent muscle lesion [4]. Mair et al. found that though leg muscles are injured at the same length regardless of fatigue, fatigued muscles are able to absorb less energy before reaching the degree of stretch that causes injuries [5]. Opar et al. cited fatigue among a list of risk factors for hamstring strain injuries (HSIs) [6]. This is especially relevant to research concerning the NFL because lower extremity injuries, in particular, account for 50% of all injuries [7].

Injury risk and its relationship to an athlete's rest has also been researched in many other professional sports. Dupont et al. studied injury rates among elite soccer players in the Union of European Football Associations Champions League and found that decreased recovery duration appeared sufficient to maintain a level of physical performance but was associated with an increased injury rate [2]. However, no academic study has investigated the effects of shorter recovery durations specifically for players participating in Thursday Night Football within the National Football League.

The aim of this study was to determine whether shortened recovery periods between regularly scheduled Sunday NFL games and Thursday Night Football games significantly increased the incidence of injury in NFL players. With growing concerns regarding player safety within the sport of football at all levels, it is vital that an attempt is made to objectively measure injury risk increases.

Methods

Official NFL injury reports are published each week during the NFL preseason, regular season, and playoffs, by every NFL team – on their respective websites, as well as league-wide websites [8]. Injury reports contain detailed information about each injured NFL player's diagnosis. The information published in injury reports includes 'player name, position, anatomical area injured

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(ankle, knee, hip, shoulder, toe, head, etc.), the practice status of a player during the week (did not practice, limited participation, full participation) and game status (probable, doubtful, out)' [7]. NFL teams are 'required to report the level of participation in practice of any player hampered by an injury' [7]. The NFL regular season consists of 16 games played over a 17-week period (each team gets one bye week). Injury reports are published prior to each NFL game; thus, the information detailed in each week's injury report reflects injuries sustained by players in the previous week's game. Injury reports do not comment on the context of a player's injury (ie. game vs practice); however, the majority of injuries suffered by NFL players during a season occur during games [9–12]. Players who are severely injured may be placed on the injured reserve (IR) list rather than the weekly injury report. The majority of players on IR will be ineligible to return to play during the given season, but NFL rules allow each team to reactivate one player from this list after they have been inactive for eight weeks. Injured reserve lists are also available online for each team and are updated daily [7].

For this study, injury report and injury reserve data were collected for every NFL player, on all 32 NFL teams, for each week during the regular season, for the 2012–2013, 2013–2014, 2014–2015, 2015–2016, 2016–2017 seasons (Table 1). Injuries were defined as any reported injury, not previously documented in an injury report prior to the said injury.

When analyzing the injury reports the authors were not able to discern whether or not an injury occurred during a practice or game; but because the vast majority of NFL injuries occur during games) we assumed that all injuries listed on the injury report occurred during games. This assumption clearly comes with limitations but it allowed us to estimate the incidence of all-cause injury in NFL players during Thursday Night Football games during the 2012–2013, 2013–2014, 2014–2015, 2015–2016, 2016–2017 seasons [9–11].

Injury incidence rates were reported as the number of injuries per 1000 athletic exposures with 95% confidence intervals. The number of athletic exposures was equal to the sum of the total number of NFL regular season games played on Saturday, Sunday and Monday (weekend games) or on Thursday. The 16 games taking place on Saturdays were included with Sunday and Monday games due to their proximity to normal Sunday games with respect to days of rest. Relative risk was calculated using the injury rate per 1000 athletic exposures for weekend NFL games vs. Thursday Night Football games with a 95% confidence interval. The statistical analysis for this study was performed using Stata Data Analysis and Statistical Software. All data included in this study were available in the public domain.

Results

During the 2012–2013, 2013–2014, 2014–2015, 2015–2016, 2016–2017 NFL regular seasons 1280 games were played. During the 2012–2013 NFL regular season 13 Thursday Night Football games were played, in 2013–2014, 2014–2015, 2015–2016, 2016–2017 that number increased to 14 games. Thus, there were 69 Thursday Night Football games between 2012 and 2017. There were 8480 players included in this study. A total of 9,621 all-cause injuries were identified over the study period. Of the 9,621 all-cause injuries 9,202 of those injuries occurred during NFL weekend games, while 419 all-cause injuries occurred during Thursday Night Football games.

The all-cause injury rate calculated during NFL weekend games was 7,598 per 1000 athletic exposures (Table 1). The calculated all-cause injury rate during Thursday Night Football games was 6,072 per 1000 athletic exposures (Table 1). The calculated relative risk of injury during Thursday Night Football games compared to weekend football games was 0.97 (95% CI, 0.95–0.98).

Discussion

Football has the highest injury rate among popular American sports [12]. Football injuries, when compared with other contact sports injuries, have been found more likely to require surgery and to end an athlete's season or career [13–15]. This study showed an injury rate of 7,598 injuries per 1000 athletic exposures for weekend games; equivalent to roughly 7.54 players injured per game. The high rate of injury amongst football players likely has contributed to the growing concerns regarding player safety within the sport of football at every level. Thus, it is vital that risk mitigation strategies be implemented in the NFL.

Prominent NFL players have publicly opposed playing Thursday night football games citing concern for increased risk of injury [1]. However, this study reported a slightly lower risk of injury during an NFL Thursday night football game compared to a weekend game with an absolute risk reduction of 3%. The result was statistically significant; thus, the rate of injury during Thursday night football games is likely less than the rate of injury during weekend football games. This result was not expected and contradicts the current literature and concerns of many players that a lack of proper recovery time will increase an athlete's risk of injury [2–6,16,17].

Table 1. Injury incidence rate in NFL players during 2012–2017.

Year	Weekend Games	Weekend Injuries	Weekend Injuries per 1000 Athletic Exposures	Thursday Games	Thursday Injuries	Thursday Injuries per 1000 Athletic Exposures
2012–2013	243	2,096	8,625	13	107	8,230
2013–2014	242	2,000	8,264	14	81	5,785
2014–2015	242	1,815	7,500	14	70	5,000
2015–2016	242	1,694	7,000	14	84	6,000
2016–2017	242	1,597	6,599	14	77	5,500
Total	1211	9,201	7,598	69	419	6,072

Weekend games include Saturday, Sunday, and Monday games. One NFL game equals one athletic exposure.

Furthermore, there is no consensus in the literature regarding an appropriate length of recovery time following a football game that would decrease the risk of injury. Practice may not share the same burden of fatigue as a game, but clearly, football players are able to practice frequently without increasing the number of injuries. It is apparent that three days may be sufficient recovery time between football games, but this remains a topic requiring further investigation.

There are limitations to this study. The lack of long-term outcomes makes it difficult to conclude that the injury rate during Thursday night football games was no different than the injury rate during weekend football games. Also, NFL injury reports lack data specific information such as the type of injury and treatment, so the type of injury and management from data on how many games were missed and whether players were placed on the IR is lacking. Injury prevention level may also have been variable depending on each athlete's position, age, and skill level. Further investigation must be performed to determine the long-term outcomes of length of recovery and injury rates surrounding Thursday Night Football.

Conclusion

During the five seasons between 2012 and 2017, the rate of injury during Thursday night football games was less than the rate of injury during Saturday, Sunday, and Monday night games, despite the lack of additional recovery time. Eliminating Thursday Night Football is unlikely to improve the statistical injury rates among NFL players. The results of our short-term analysis show a statistical decrease in injury rates between weekend and Thursday night NFL games. Further data and larger sample sizes of Thursday night games would be helpful in drawing more concrete conclusions.

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Declaration of interest

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